

Aware in Medicine

Making medical knowledge accessible for all.



Never Too Late: A Guide To Reverse Type 2 Diabetes

Frequent thirst. Hunger. Urination. Craving sugar. If your body is trying to talk to you, it's time to listen.

Author: Olivia Wang

Date Published: 01.09.26

Reading Time: [10 minutes]



Abstract

Most people view diabetes as "eating too much sugar" or "always taking insulin injections," but it is much more than that. The world doesn't understand that more than 830 million people worldwide are affected by type 2 diabetes, yet more than 50% of them are still left undiagnosed. People should gain more knowledge about this condition to prevent and treat it. That is exactly what this article will give readers: a guide to understanding type 2 diabetes, its causes, symptoms, diagnosis, treatment and next steps. Stop wondering what to do about diabetes and start being the answer.

Introduction

Type 2 diabetes affects more than 830 million people worldwide and 50% of them remain still undiagnosed. According to the research (Mayo Clinic, section 1) we can determine that type 2 diabetes is a chronic disease that alters the procedure of breaking down glucose, a type of sugar that gives energy to our cells. For people not affected by diabetes, the glucose is broken down and enters the bloodstream, increasing the sugar level in the blood. To compensate for the rise, insulin is added, an essential hormone produced by the pancreas that stimulates glucose to enter cells for storage and energy (Cleveland Clinic, section 2).

For those with type 2 diabetes, this process of digesting glucose doesn't function as well and requires external regulations. It is triggered by two main factors: the inability of the pancreas to produce enough insulin and the body's weak response to it. Becoming less sensitive to the insulin signal is known as becoming insulin resistant.

According to research (Global Self Healthcare Education and Research, section 5), as a result of the weakened signal from the target tissues in the skeletal muscles, liver, and fats, the body tells the pancreas to produce excessive insulin, which builds up in the blood. The weakened signal and its receptor, a special protein on the cell surface unlocking specific structures, act as a key trying to open a coated lock. Since the pancreas wants the glucose to enter the coated cells anyways, it decides to force more insulin against the cells. Over time, the pancreas loses the ability to produce this hormone and requires insulin injections.

At first, there are minor symptoms that often go unnoticed: weight loss, blurred vision, frequent thirst, urination, hunger, and craving sugar. Your body is trying to tell you that there is a problem with your pancreas and that you should bring attention to it. If you ignore those symptoms, massive amounts of glucose may start to build up in the bloodstream, elevating the chance for future life-threatening disease, like cardiovascular and neurological complications. That's why understanding this disease can help you control your blood sugar level and prevent fatal damage.

Causes

1 in 3 adults are most likely to have diabetes during their lifetime, according to (UW Medicine, section 1). Although type 2 diabetes is partly genetic, it is more commonly

caused by lifestyle factors and age. Genetics play a role in the family history of type 2 diabetes, obesity or the thousands of gene variants that affect the processing of glucose. However, this disease weighs more on age, gender and lifestyle, like diet, stress level and living environment. A 70 year old man who works in a factory and eats BeaverTails every night statistically has a higher risk of type 2 diabetes than a 20 year old female who works in a hospital and cooks home-made food.

Risk Factors

Other than age, gender and lifestyle, statistics show prominent risk factors for this disease.

- Being prediabetic
- Having high cholesterol
- Having a high blood pressure
- Having a family history of diabetes
- Being Indigenous, African, East Asian, South Asian
- Living as a middle-income class (11%) vs. a low-income class (7%)

What Brings You a Step Closer...

According to a Harvard Health study, having fried foods every day increases the risk of type 2 diabetes by 50%, as it leads to the three main causes of diabetes: weight gain, high cholesterol and high blood pressure. Eating foods high in saturated fat, like milk or red meat also enhances the chance of type 2 diabetes, playing a part in insulin resistance. Worst of them all, sugary foods or drinks, causing glucose to rapidly spike and your pancreas to release big amounts of insulin. This will eventually lead to the loss of the ability for this organ to produce insulin.

Believe it or not, high levels of stress can also increase the risk of type 2 diabetes. When a person is stressed, cortisol and adrenaline are released, which are the body's primary stress hormones. Glucose is then dumped into the bloodstream, making you feel the burst of energy.

People who live in an atmosphere with little air pollution and a lot of green space will more likely walk and do physical activity. A person living close to a supermarket is also more probable to cook organic food at home.

Symptoms & Different Stages of Type 2 Diabetes

Insulin resistance: Developing insulin resistance is the first stage of type 2 diabetes and will most likely not result in any change. As one of the main causes of type 2 diabetes, the pancreas doesn't pick up a strong enough signal that directs it to produce a certain amount of insulin. At this stage, the pancreas is already starting to act on unusual activity and blood sugar levels are slowly going to rise.

Prediabetes: At this stage, people may notice mild symptoms, like blurred vision, weight loss, frequent thirst, hunger, urination and craving sugar, or none at all. This explains why 73% of patients are left undiagnosed and could have been living with this disease for many years already. If you feel any of these symptoms, it's best to check in with a doctor to diagnose whether or not you are prediabetic to create a medical plan. The only way to find out is by taking a blood test called the Hemoglobin A1C test. According to research (Gunther, section 2; UW Medicine), blood sugar levels residing between 100 milligrams/deciliter (mg/dL) and 125 mg/dL means that someone is prediabetic and may face future complications with type 2 diabetes.

Compared to the later stages, this condition is reversible and if people become more aware about their lifestyle, they can become non-prediabetic. However, once you no longer show a high blood sugar level, you should maintain that level, because if you were prediabetic, there is a higher chance that you can become one again.

Type 2 Diabetes: As a person enters the earlier stages of type 2 diabetes, these symptoms get more noticeable, as their blood sugar levels rise above 126 mg/dL. Adding on to the symptoms listed in prediabetes, you may also start experiencing tingling or numbness in the feet or hands and wounds that take a long time to heal. Now, the pancreas is exhausted due to insulin resistance and starts to produce less and less insulin.

Type 2 Diabetes with Complications: However, when this disease progresses to the later stages, it is too late. You have had type 2 diabetes for many years now and you still haven't got it treated. Based on (Govindarajan Venguidesvarane et al., section 1), this situation accounts for over 80% of the population in a Rural Health Center and most of them got diagnosed at this stage. This condition is responsible for many complications, like blindness, since the blood

vessels in the eyes start to swell up due to the excess glucose. This damages the vessel walls and leaks fluid and blood into the eye.

The most common complications are heart attacks or strokes: when the excess glucose mixes with fat and cholesterol in the blood and forms plaques. This blocks the blood from flowing freely and eventually stops the circulation.

Nevertheless, the most rare but probable, is when the body is no longer able to fight infection due to the damage done to white blood cells. This causes people to have to amputate their wounds in order to prevent life-threatening infection from spreading.

Note: Symptoms vary widely between individuals; readers should not self-diagnose based on this list. If someone suspects they have this condition, they should consult a healthcare provider.

Diagnosis

Diagnosis is your lifesaver. Type 2 diabetes is known as a "silent killer," which is why you have to get tested with a Hemoglobin A1C test every few months at the clinic. Listed below are the ranges of normal, prediabetic and diabetic blood sugar concentration.

- Normal = less or equal to 100 mg/dL
- Prediabetic = 100 mg/dL -125 mg/dL
- Diabetic = more or equal to 126 mg/dL

This could also be done daily using a glucometer at home, a portable medical device to measure blood sugar level. First, you would insert the test strip into the glucometer. Next, use the lancet, or tiny blade, to prick the side of your finger and collect a droplet of blood onto the tip of the strip. Read the number on the glucometer and assess your blood sugar concentration.

Other tests also exist to identify type 2 diabetes, such as the diabetic foot examination. When a person has type 2 diabetes, they often lose the ability to feel the endings of their nerves. An expert will first use a plastic thread and put pressure to different parts of your foot to make sure that the nerve endings are still functioning. Then, they will put a 128 Hertz (Hz) tuning fork, a medical tool, onto the corner of both feet to confirm that you can feel the vibration.

Before being diagnosed with type 2 diabetes, doctors might assess the patient's medical history to bring out an accurate diagnosis and make sure that the context fits in. They also consider their family history, the symptoms being experienced and any physical abnormalities, like obesity, hypertension and eye hemorrhages. Their immunization history might as well be taken into account to analyze the level of strength of the patient's immune system, as diabetes highly compromises it.

Treatment

If a person is diagnosed with type 2 diabetes, doctors might offer different treatment options to avoid future complications. Depending on certain stages, there will be specific procedures to take. However, regardless of the stage, a person should try to treat the following conditions if experienced:

Hypertension: According to research (Orrange, section 3), hypertension could be treated with a daily intake of prescription medications. One type of medication includes the angiotensin-converting enzyme inhibitor or ACE inhibitor. Its function is to prevent the formation of angiotensin II, a hormone that increases blood pressure.

Based on research (NIH, section 2; Orrange), modifying your diet can also be very helpful, such as following the DASH diet, consisting of plenty of vegetables, fruits, whole grains, low-fat dairy, vegetable oils and lean meats. Limiting alcoholic beverages, sugary drinks and red meats is recommended.

Plaque Buildup: According to research (Harvard Health Publishing, section 3), doctors often prescribe statins, medications that block the liver enzyme from producing cholesterol. Cholesterol is the one that if consumed excessively, builds up in the inner side of the arteries to create plaques. As a result, you may experience shortness of breath or chest pain.

In addition, there is a possibility that the plaque could rupture, causing a heart attack. When a plaque ruptures, the substance filled with cholesterol, calcium and white blood cells seeps into the bloodstream and the body starts to create blood clots. This cuts off the circulation to vital organs, like the heart, brain and lungs.

Emphasize: Treatment plans are highly individualized; readers should always consult their personal physician before starting or modifying treatment.

Management

For individuals with insulin resistance or that are prediabetic, there are a few steps that can control their blood sugar level and prevent the occurrence of this condition.

As mentioned before, these include maintaining a healthy body weight, since it minimizes the blockage of the "lock" of the cell, allowing glucose to enter the cell for storage.

Maintaining a healthy diet was also very important. Minimizing foods high in sugar, cholesterol and saturated/trans fats as well as processed foods can lower the chance for cardiovascular and neurological conditions.

And most importantly, adapting lifestyle factors and increasing physical activity. This will lower blood sugar levels with the help of proper digestion of glucose and its efficient conversion to energy.

Richard's Story

"I'm Richard King and I've lived comfortably with type 2 diabetes for over 20 years," King states, according to (NKC Health).

When he was first diagnosed with type 2 diabetes, his mind immediately went through all the possible risks and dangers that he could encounter. The main dangers being heart attack or stroke, kidney failure and nerve damage. From then on, he decided that he wasn't going to let diabetes determine his fate and went on to adapt his lifestyle and diet. After increasing physical activity and drastically changing eating habits nearly every day, he lost 30 pounds the next time he visited his physician, Dr. Dunbar.

King wanted to be a healthy individual, despite his condition of type 2 diabetes and he sends a message to inspire those experiencing any stage of diabetes: if you don't feel well, don't ignore these indications and go to the doctor. King consulting with Dr. Dunbar as one of the best decisions that he has ever made during this journey. She

stepped in King's shoes and understood his health concerns and gave in her thoughts of how to manage his condition for the future.

One piece of advice that he gave to himself was to always have a clear ruler in his head telling you to constantly change your diet and lifestyle factors. No matter how strong the urge to experience what you had before is, the urge to become a healthier individual should be stronger.

This led him to his final message which was to maintain your blood sugar level on a regular basis by checking with a glucometer. This will allow you to understand your current stage and figure out your next steps onwards.

Now What?

In the current world, we are able to understand type 2 diabetes, diagnose this condition, as well as treat it, and maintain a healthy condition afterwards. According to research (Shomali, section 1), in future years, scientists are looking at an increase of type 2 diabetes cases as people become more inactive through the convenience of technology. Even though physicians and experts continuously remind patients to maintain a healthy lifestyle, patients never seem to follow their instructions until they experience the consequences. This means that diabetes will slowly become part of our lives and the research that is being conducted is essential for the future.

"Having diabetes might not be your choice, but managing it is," (Advocators). Type 2 diabetes is a treatable and manageable disease that is empowered by your own hands. This disease is one that you inevitably live with your entire life and it is totally up to you whether you want to change that. Becoming a healthier person and adapting your lifestyle is a hard decision, but it is one that your later self will thank your current self for. Be that person: see a doctor, diagnose your disease and get treatment. Remember, "diabetes may be tough, but you are tougher," (Advocators).

References

All claims in this article are supported by evidence-based sources.

Your References:

1. American Diabetes Association. « Understanding Diabetes Diagnosis ». *American Diabetes Association*, American Diabetes Association, 2023, <https://diabetes.org/about-diabetes/diagnosis>.
2. Areosa Sastre, Almudena, et al. « Effect of the Treatment of Type 2 Diabetes Mellitus on the Development of Cognitive Impairment and Dementia ». *Cochrane Database of Systematic Reviews*, 15 juin 2017, https://www.cochrane.org/CD003804/DEMENTIA_effect-treatment-type-2-diabetes-mellitus-development-cognitive-impairment-and-dementia, 10.1002/14651858.cd003804.pub2.
3. CDC. « Vision Loss and Diabetes ». *Diabetes*, 13 mai 2024, <https://www.cdc.gov/diabetes/diabetes-complications/diabetes-and-vision-loss.html>. Consulté le 21 août 2026.
4. Cleveland Clinic. « Insulin ». 17 janv. 2024.
5. Dendup, Tashi, et al. « Environmental Risk Factors for Developing Type 2 Diabetes Mellitus: A Systematic Review ». *International Journal of Environmental Research and Public Health*, vol. 15, n° 1, 2018, p. 78. *PubMed Central*, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5800177/>, 10.3390/ijerph15010078. Consulté le 31 juill. 2026.
6. Galicia-Garcia, Unai, et al. « Pathophysiology of Type 2 Diabetes Mellitus ». *International Journal of Molecular Sciences*, vol. 21, n° 17, 30 août 2020, p. 1-34, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7503727/>.
7. Government of Canada. « Diabetes: Prevention and Risk Factors - Canada.ca ». 2025.
8. Govindarajan Venguidesvarane, Akila, et al. « Prevalence of Vascular Complications Among Type 2 Diabetic Patients in a Rural Health Center in South India ». *Journal of Primary Care & Community Health*, vol. 11, janv. 2020, p. 215013272095996, 10.1177/2150132720959962.
9. GP Evidence. « Type 2 Diabetes Mellitus GP Evidence ». *GP Evidence*, 2023, <https://gpevidence.org/conditions/test/>.
10. Gunther, Anna. « What Is the Normal Range for Blood Sugar? ». 2024.
11. Harvard Health Publishing. « Can We Reduce Vascular Plaque Buildup? - Harvard Health ». Harvard Health, 3 mai 2019.
12. Henning, Robert J. « Type-2 Diabetes Mellitus and Cardiovascular Disease ». *Future Cardiology*, vol. 14, n° 6, nov. 2018, p. 491-509, 10.2217/fca-2018-0045.
13. Integris Health. « Food Choices That Impact Your Risk of Type 2 Diabetes ». 6 janv. 2023.
14. Mayo Clinic. « Type 2 Diabetes ». 2025.
15. NIH. « DASH Diet to Lower High Blood Pressure: MedlinePlus Medical Encyclopedia ». 2013.
16. NKC Health. *Live the Sweet Life With Type 2 Diabetes | Patient Story | NKC Health*. YouTube, 5 sept. 2018.
17. Orrange, Sharon. « Choosing Blood Pressure Medication: Which Is Best for You? ». *GoodRx*, 31 janv. 2017, <https://www.goodrx.com/conditions/hypertension/choosing-your-blood-pressure-medication-what-type-is-best?srsId=AfmBOoqcpocrXXHZYxcAH0toAyYSp7RmKwzLknO7dvnXOWoqymtJG-3I>. Consulté le 31 juill. 2026.

18. Romesh Khardori, MD, PhD, FACP. « Type 2 Diabetes Mellitus Clinical Presentation: History, Physical Examination ». *Medscape.Com*, Medscape, 23 déc. 2024, <https://emedicine.medscape.com/article/117853-clinical#showall>.
 19. Shomali, Mansur. « Diabetes Treatment in 2025: Can Scientific Advances Keep Pace With Prevalence? ». *Therapeutic Advances in Endocrinology and Metabolism*, vol. 3, n° 5, oct. 2012, p. 163-173, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3498849/>, 10.1177/2042018812465639. Consulté le 31 juill. 2026.
 20. The Lancet Diabetes & Endocrinology. « Aligning Diabetes Surveillance with Clinical Reality ». *The Lancet Diabetes & Endocrinology*, vol. 14, n° 2, févr. 2026, p. 93, 10.1016/s2213-8587(25)00404-8. Consulté le 9 mars 2026.
 21. UW Medicine. « 1 In 3 U.S. Adults Has Prediabetes, Many Unknowingly | Department of Medicine News ». *Uw.Edu*, 2023, <https://mednews.uw.edu/news/prediabetes>. Consulté le 21 août 2026.
 22. Williams, Angela. « How Sleep and Stress Affect Blood Sugar | University Health | University Health ». 2025.
 23. World Health Organization. « Diabetes ». *World Health Organization*, 14 nov. 2024, <https://www.who.int/news-room/fact-sheets/detail/diabetes>. Consulté le 21 août 2026.
-

Aware in Medicine (AiM)

Making medical knowledge accessible for all.

Aware in Medicine is a youth-led non-profit learning hub dedicated to improving health literacy by breaking down complex medical conditions into clear, research-backed explanations for patients, students, and families.

awareinmedicine.org | awareinmedicine@gmail.com

© 2026 Aware in Medicine. All rights reserved.